

PRANAV KISHAN T Y

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Computer Science undergraduate experienced in full-stack web development, backend engineering. Proficient in Python, Java, SQL, and TypeScript. Skilled in building real-time web applications, engineering secure APIs, and optimizing database architectures.

EDUCATION

Bachelor of Technology - Computer Science and Engineering

2023-2027

Amrita Vishwa Vidyapeetham - Coimbatore, India

CGPA - 8.66

- **Relevant Coursework:** Data Structures and Algorithms, Object Oriented Programming, Operating Systems, Computer Networks, Database Systems, Software Engineering, Machine Learning, Neural Networks and Deep Learning

TECHNICAL SKILLS

- **Languages:** Python, SQL, Java, C/C++, JavaScript, TypeScript
- **Databases & Cloud:** PostgreSQL, MySQL, MongoDB, SQLite, Google Cloud Platform
- **Backend & Systems:** FastAPI, Node.js, Express.js, REST APIs
- **Tools & Platforms:** Git, GitHub Actions, Docker, Linux, Bash
- **Security:** AES/RSA Encryption, Digital Signatures, RBAC, Zero-Knowledge Principles

PROJECTS

- **Netwatchpy** | [Repo](#) | [PyPI](#) | Python, SQLite, GitHub Actions
A cross-platform, installable TUI for real-time network usage monitoring.
 - Refactored SQLite schema from text strings to **Unix Epoch integers**, reducing range query latency by **~40%** via optimized **B-Tree indexing**.
 - Engineered a **fault-tolerant SQLite WAL architecture** to process real-time telemetry and render 24-hour bandwidth metrics **without database locks**.
 - Established a GitHub Actions CI/CD pipeline to automate cross-platform binary builds and PyPI releases, accumulating over 11,000 total downloads.
- **InfoStuffs** | [Repo](#) | [Live](#) | [Blog](#) | React.js, MongoDB, Express.js, Node.js, Vercel, Docker
Zero-Knowledge personal information manager ensuring end-to-end privacy via client-side cryptography.
 - Engineered an **end-to-end encrypted, zero-knowledge vault** using client-side AES-GCM, Web Crypto API, and PBKDF2 to guarantee **absolute data confidentiality**.
 - Architected a **zero-cost backend** combining a **persistent Node.js Docker container** on Render to eliminate database cold starts, alongside **Vercel serverless functions** for automated tasks.
- **SurplusSync** | [Repo](#) | [Live](#) | NestJS, React.js, PostgreSQL, Docker, GitHub Actions, Redis, Socket.io
Real-Time Food Redistribution Platform.
 - Acted as **Backend/DevOps & Scrum Master**, managing **Agile workflows** and cross-functional team sprints.
 - Set up a **continuous integration (CI) pipeline** using **GitHub Actions** to automate testing and build integrity for a **containerized monorepo**.
 - Implemented **real-time event infrastructure** using **Socket.io** for live alerts and integrated **Redis** for caching, alongside automated **PostgreSQL database backups**.

EXPERIENCE

- **Backend Developer - Magilam Foods (Client Project)** **February 2026 - Present**
 - Developed an inventory management backend featuring automatic stock deduction and low-stock alerts.
 - Ensured strict database consistency and seamless data synchronization across complex order workflows.
- **Intel IoT Club - Full-Stack Developer** **November 2025 - Present**
 - Resolved critical **server-client rendering mismatches** (hydration errors) to ensure deterministic application behavior and visual stability on the official club platform.
 - Implemented **modular components** for the EasyEPFO payment interface and **refactored core website layouts** to improve visual consistency.
- **ACM Student Chapter - Member** **May 2025 - Present**
 - Engineered a **strictly typed, reusable UI component** library using React and TypeScript, resolving **cross-device responsiveness issues** and standardizing the design system across the Chapter and SOC portal.

ACHIEVEMENTS

- Our team **secured 39th place** in the **Puch.AI Hackathon** with over 25,000 participants.
- **Selected** for the **Visteon Scholar Program**, a competitive mentorship initiative for engineering undergraduates.

CERTIFICATIONS AND COURSES

- IETE Docker Workshop | [CERTIFICATE](#)
- Google Cloud - Introduction to GenAI | [CERTIFICATE](#)
- Google Cloud - Prompt Design in Vertex AI | [CERTIFICATE](#)
- Introduction to Distributed SQL and CockroachDB, Cockroach University | [CERTIFICATE](#)